

SANTIAGO LLORENS

Scholar Profile

<https://scholar.google.com/citations?user=wMDRPZYAAAAJ&hl=en>

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Education

2022-2025	Ph.D. in Physics, Quantum Information Group, Department of Physics, Autonomous University of Barcelona: New techniques for verification and identification of quantum states and processes
2020-2021	Master's Degree in Quantum Science and Technology, Basque Country University The geometry of composite systems: Bloch ball analog for two qubits
2015-2020	Bachelor's Degree in Physics and Bachelor's Degree in Chemistry, Autonomous University of Barcelona

Laboral Experience

2026-Present	Post-doctoral researcher, Division of Quantum Information, Faculty of Mathematics, Physics and Informatics, University of Gdańsk,
2025	Post-doctoral researcher, Quantum Information Group, Department of Physics, Autonomous University of Barcelona
2022-2025	Pre-doctoral researcher, Quantum Information Group, Department of Physics, Autonomous University of Barcelona
2020-2021	Physics Department, Basque Country University Master's Thesis on Bloch Representation on Finite-Dimensional Composite Systems

Publications

- 2026** J. Stempin, G. Anglès-Munné, S. Llorens, and F. Huber *Combining moment matrices, symmetric extension, and Lovász theta: Φ_{E8} is entangled*, arXiv pre-print: arXiv:2605.13832
- 2026** S. Llorens, A. Diebra, M. Sedlák, and Ramon Muñoz-Tapia *Exact identification of unknown unitary processes*, arXiv pre-print: arXiv:2605.04981
- 2025** A. Diebra, S. Llorens, D. González-Loziga, A. Rico, J. Calsamiglia, M. Hillery and E. Bagan *Quantum Advantage in Identifying the Parity of Permutations with Certainty*, Physical Review Letters **135**(26), 260603)
- 2025** A. Diebra, S. Llorens, G. Sentís, R. Muñoz-Tapia and E. Bagan *Quantum state exclusion for group-generated ensembles of pure states*, Physical Review Research **8**(1), L012001
- 2025** S. Llorens, W. González, G. Sentís, J. Calsamiglia, R. Muñoz-Tapia and E. Bagan *Quantum edge detection*, Quantum **9**, 1687 (2025)
- 2024** M. Skotiniotis, S. Llorens, R. Hotz, J. Calsamiglia, R. Muñoz-Tapia *Identification of malfunctioning quantum devices*, Physical Review Research **6**, 033329(2024)
- 2024** S. Llorens, G. Sentís and R. Muñoz-Tapia *Quantum multi-anomaly detection* Quantum **8**, 1452 (2024)
- 2024** S. Morelli, S. Llorens and J. Siewert *Exploring Imaginary Coordinates: Disparity in the Shape of Quantum State Space in Even and Odd Dimensions*, arXiv pre-print: arxiv:2404.15179

Teaching responsibilities

2024/2025 Accredited hours: 8	Co-director of Bachelor's thesis, Physics bachelor's degree, UAB, <i>Lower bounds for quantum state exclusion</i>
2024/2025 Accredited hours: 88	Teaching assistant: Quantum Physics II, Physics bachelor's degree, UAB
2023/2024 Accredited hours: 8	Co-director of Bachelor's thesis, Physics bachelor's degree, UAB, <i>Association schemes for quantum state identification</i>
2023/2024 Accredited hours: 93	Teaching assistant: Quantum Physics II, Physics bachelor's degree, UAB
2023/2024 Accredited hours: 16	Teaching assistant: Current topics in Science, Physics bachelor's degree, UAB
2022/2023 Accredited hours: 92	Teaching assistant: Quantum Physics II, Physics bachelor's degree, UAB
2022/2023 Accredited hours: 45	Teaching assistant: Physics for Geology, Geology bachelor's degree, UAB
2021/2022 Accredited hours: 44	Teaching assistant: Physics for Geology, Geology bachelor's degree, UAB

Conferences

April 2026 Attendance	17th KCIK-ICTQT Symposium on Quantum Information.
October 2025 Contributed Talk:	Quantum Information in Spain ICE-10, Valencia, Spain. <i>Learning from the unknown: Universal Quantum State Discrimination</i>
October 2025 Poster	XL Trobades Científiques de la Mediterrània-Josep Miquel Vidal: Ciència i Tecnologies Quàntiques a la Mediterrània, Maó, Spain
February 2025 Organizer	VII Pyrenees Winter School in Quantum Information and Quantum Science, Setcases, Spain
June 2024 Contributed Talk	19th Central European Quantum Information Processing Workshop, Skalica, Slovakia, <i>Quantum multi-anomaly detection</i>
February 2023 Contributed Talk	VI Pyrenees Winter School in Quantum Information, Setcases, Spain. <i>Quantum edge detection</i>
September 2022 Contributed Talk	Deutschen Physikalischen Gesellschaft, Regensburg, Germany. <i>Hilbert-Schmidt geometry of two-qubit correlations</i>
May 2022 Contributed Talk	Quantum Information in Spain ICE-7, Granada, Spain. <i>The Geometry of Composite Systems: Bloch-Ball Analog for Two Qubits</i>

Journal and conference reviewer

Journal	Quantum Physical Review A
Conference	Theory of Quantum Computation IEEE International Symposium on Information Theory

Languages and Other Skills

Languages	Spanish, Catalan: native speaker English
Programming	Mathematica, Matlab, Python, Bash, C++, AWK
Experiences	Participation and completion of IBM Quantum Challenge 2020.